VECTOR Model Specification — Bipartite Assortative Formation with Finite-Capacity Pools

Last synced: 2026-08-10
Status: Working specification — baseline MBB paradigm
Purpose: Record the current notation, assumptions, timing, and implementation locks for the minimal cross-domain assortative formation model.

1. Core Modeling Idea

The model is a **bipartite formation process**. Individual agents attach to team/pool nodes rather than directly to other individual nodes.
The sole baseline formation mechanism is **similarity attraction**: an incoming agent is more likely to affiliate with a team whose **lagged centroid** is similar to the agent's own performance/ability value.

$$\text{prior roster} \to \text{lagged team centroid} \to \text{similarity-based assignment} \to \text{new roster} \to \text{new performance} \to \text{new centroid.}$$

The centroid is not a permanent team type or exogenous target. It is an endogenous state generated by the previous roster.

2. Network Structure and Performance

Individuals are indexed by $i = 1, \dots, N$ and teams/pools by $j = 1, \dots, J$.
Each active individual belongs to exactly one team at a given snapshot.
Let

$$A_{i,t}$$

denote individual i's performance/ability at the end of period t. Performance is time-varying:

$$A_{i,0}, A_{i,1}, A_{i,2}, \dots$$

3. Team / Pool Centroid

Use

$$\boxed{\mu_{j,t}}$$

for the **team centroid**, or **pool centroid** in domain-general language.
For the one-dimensional performance attribute:

$$\boxed{\mu_{j,t} = \frac{1}{n_{j,t}} \sum_{i \in T_{j,t}} A_{i,t}}$$

where $T_{j,t}$ is team j's roster and $n_{j,t} = |T_{j,t}|$.
Centroid replaces PPM because PPM already means **points per minute** in the basketball work.
Fitness is intentionally avoided. In the Bianconi–Barabási fitness model, fitness is an intrinsic node characteristic affecting competition for links. Here, $\mu_{j,t}$ is **endogenous, lagged, and compositional**.

4. Frozen Lagged Centroid

Assignments for period t use

$$\boxed{\mu_{j,t-1}}.$$

This value remains **frozen for the entire assignment round**. It is not recalculated as new players arrive.
After period t, new individual performances are realized and the new centroid is calculated:

$$\mu_{j,t} = \frac{1}{n_{j,t}} \sum_{i \in T_{j,t}} A_{i,t}.$$

Thus:

$$\boxed{\mu_{j,t-1} \to \text{formation at } t \to \mu_{j,t} \to \text{formation at } t+1.}$$

This prevents an arriving player from altering the same team signal that attracted that player.

5. Team Capacity

Let C_j denote team j's carrying capacity.

The general model allows C_j to vary, but the baseline imposes

$$\boxed{C_j = C \qquad \forall j.}$$

For the initial NCAA men's basketball implementation:

$$\boxed{C = 15.}$$

Let $n_{j,t}^{(s)}$ denote team j's occupancy after assignment step s.
When

$$n_{j,t}^{(s)} = C_j,$$

team j is full and leaves the remaining choice set.

6. Feasible Team Set

Define

$$\boxed{\mathcal{F}_t^{(s)} = \{j : n_{j,t}^{(s)} < C_j\}.}$$

This contains **all teams with an open roster position** when the s-th assignment is made.

The baseline model has no artificial conference, geographic, prestige, recruiting, or other consideration-set restrictions. Every non-full team is available.

Capacity changes the **feasible choice set**, not the definition of similarity.

7. Individual–Team Distance and Similarity

Let A_i^* denote the performance information available when individual i is assigned.

For an incumbent/mover with prior-period performance:

$$A_i^* = A_{i,t-1}.$$

For a brand-new entrant:

$$A_i^* = A_{i,t}^{\text{entry}}.$$

Define distance from candidate team j's lagged centroid:

$$\boxed{d_{ij,t} = |A_i^* - \mu_{j,t-1}|.}$$

If A is normalized appropriately, a Quayle-style similarity score may be written:

$$\boxed{x_{ij,t} = 1 - d_{ij,t}.}$$

Here, j indexes a **team node**, not another player node.

The conceptual translation is:

$$\text{Quayle-style individual } i \to \text{similar individual } j$$

becoming

$$\boxed{\text{individual } i \to \text{similar team } j.}$$

8. Assortative Attraction

Let

$$\boxed{\rho}$$

denote the strength of **assortative attraction**.

Define the similarity weight:

$$\boxed{w_{ij,t} = e^{-\rho d_{ij,t}}.}$$

At $\rho = 0$, every feasible team receives identical similarity weight. As ρ increases, assignment increasingly favors teams whose lagged centroid is close to the individual's performance.

The baseline model contains **no popularity or preferential-attachment term**.

Thus:

$$\boxed{\text{similarity attraction} + \text{finite capacity.}}$$

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9. Assignment Probability

For $j \in \mathcal{F}_t^{(s)}$:

$$\boxed{P(i \to j \mid t, s) = \frac{e^{-\rho d_{ij,t}}}{\sum_{l \in \mathcal{F}_t^{(s)}} e^{-\rho d_{il,t}}}.}$$

Equivalently:

$$P(i \to j \mid t, s) = \frac{e^{-\rho |A_i^* - \mu_{j,t-1}|}}{\sum_{l \in \mathcal{F}_t^{(s)}} e^{-\rho |A_i^* - \mu_{l,t-1}|}}.$$

Notation:

- i = individual currently choosing;
- j = candidate team being evaluated;
- l = dummy index over all currently feasible teams;
- $\mu_{j,t-1}$ = candidate team's frozen lagged centroid;
- $\mathcal{F}_t^{(s)}$ = all teams that still have capacity;
- ρ = assortative attraction strength.

At $\rho = 0$, every open team is equally likely. As ρ increases, probability concentrates on teams whose centroid is more similar to the individual's performance.

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10. Random Assignment Order

Players entering the assignment process are **not sorted by A**.

Let

$$\pi_t = (\pi_{1,t}, \ldots, \pi_{V_t,t})$$

be a random permutation of individuals requiring assignment in period t.

Assignments proceed sequentially according to π_t.

This matters because finite capacity makes order potentially consequential. Randomization prevents the baseline mechanism from deliberately privileging high- or low-performing entrants.

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11. Pool Tenure and System Tenure

The model ultimately distinguishes time in a current pool from time in the larger system.

Define

$$\boxed{\tau_i^P}$$

as individual i's maximum **pool tenure**, and

$$\boxed{\tau_i^S}$$

as maximum **system tenure**.

For the initial MBB paradigm:

$$\boxed{\tau_i^P = \tau_i^S = \tau = 4.}$$

After four seasons, the baseline basketball player exits the modeled system.

Later domains can separate these quantities. For example, an Army agent can complete a pool tenure, leave the current unit, remain in the system, and re-enter the same ASSIGN mechanism.
Let $a^P_{i,t}$ denote current pool-tenure age, using:

$$a^P_{i,t} \in \{0, 1, \ldots, \tau^P_i - 1\}.$$

Thus 0 is the first season and $\tau^P_i - 1$ is the final season.

12. Staggered Initial Tenure

The initial population should not consist entirely of first-period individuals.
For the MBB baseline:

$$a^P_{i,0} \sim \mathrm{DiscreteUniform}\{0, 1, 2, 3\}.$$

This creates staggered turnover immediately rather than producing artificial universal turnover after four simulated seasons.
The initialization can later be replaced with empirical tenure-age information where available.

13. Vacancies and Incoming Cohort

In the baseline MBB model, players whose tenure has not expired remain on their teams. Final-term players exit and create vacancies.
Ignoring additional attrition initially:

$$\boxed{V_t = \sum_i \mathbf{1}\left(a^P_{i,t-1} = \tau^P_i - 1\right).}$$

The incoming cohort restores league capacity:

$$\boxed{N^{\mathrm{in}}_t = V_t.}$$

With J teams and uniform capacity C:

$$\boxed{K = JC.}$$

For MBB:

$$K = 15J.$$

Equivalently, if N^{inc}_t incumbents remain:

$$N^{\mathrm{in}}_t = 15J - N^{\mathrm{inc}}_t.$$

14. New Entrant Ability Distribution

A brand-new entrant has no prior-period performance observation inside the modeled system.
Define:

$$\boxed{A^{\mathrm{entry}}_{i,t} \sim F^{\mathrm{entry}}_A.}$$

The first implementation will use an **overall ability distribution informed by empirical data** so that simulated entrants remain anchored to realistic performance values.
The exact estimator is not yet locked. Possibilities include empirical resampling or an empirically fitted distribution.

Required Distribution Diagnostic

Any simulation code that generates incoming-player talent must also plot the distribution from which those entrant values are drawn.
At minimum, code should make visible:

1. the empirical reference distribution used to construct F^{entry}_A; and
2. the realized distribution of simulated entrant draws.

This is a standing implementation requirement.

15. Empirical Initialization First

The initial implementation should begin from an **empirical system state**.
Where available, an observed season can provide:

- initial rosters;

- initial individual performance values;
- initial centroids $\mu_{j,0}$;
- initial occupancy;
- tenure-age information or an empirically informed approximation.

Thus:

$$\boxed{\mu_{j,0} = \frac{1}{n_{j,0}} \sum_{i \in T_{j,0}} A_{i,0}.}$$

This is an **initial condition**, not a persistent team target.

After initialization:

$$\mu_{j,0} \to \mu_{j,1} \to \mu_{j,2} \to \cdots$$

A later robustness phase can replace empirical initialization with synthetic initialization drawn from empirically informed distributions.

Recommended order:

$$\boxed{\text{empirical initialization} \to \text{synthetic initialization / robustness.}}$$

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16. Baseline Seasonal Timeline

For each period t:

1. Begin with known $A_{i,t-1}$, $\mu_{j,t-1}$, and $a^P_{i,t-1}$.
2. Identify players completing their final MBB term.
3. Remove those players and count vacancies.
4. Generate exactly enough entrants to restore total capacity.
5. Draw entrant abilities from F_A^{entry}.
6. Randomize the entrant assignment order.
7. For each entrant, evaluate **every currently non-full team**.
8. Sample a team using the ρ-weighted assignment probability.
9. Remove a team from $\mathcal{F}_t^{(s)}$ immediately when it reaches $C = 15$.
10. Keep every $\mu_{j,t-1}$ frozen throughout assignment.
11. Run/observe season t and obtain $A_{i,t}$.
12. Recalculate every $\mu_{j,t}$.
13. Advance tenure ages and repeat.

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17. Compact State Transition

$$\boxed{\{A_{i,t-1}, \mu_{j,t-1}, a^P_{i,t-1}\} \to \text{TURNOVER} \to \text{ASSIGN} \to \text{PERFORMANCE} \to \{A_{i,t}, \mu_{j,t}, a^P_{i,t}\}.}$$

The endogenous feedback loop is:

$$\boxed{\mu_{j,t-1} \to \text{who joins } j \to \text{composition at } t \to \mu_{j,t} \to \text{who is attracted at } t+1.}$$

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18. Notation Table

Symbol	Meaning	Baseline MBB treatment
i	individual/player index	agent
j	team/pool index	team
l	dummy index over feasible teams	all open teams
$A_{i,t}$	individual performance/ability	time-varying
$\mu_{j,t}$	team/pool centroid	endogenous mean
$d_{ij,t}$	individual–team distance	absolute distance
$x_{ij,t}$	individual–team similarity	derived from distance
ρ	assortative attraction strength	primary sorting parameter
C_j	team capacity	$C_j = C$
C	uniform capacity	15
$\mathcal{F}_t^{(s)}$	feasible team set	all non-full teams
τ_i^P	pool tenure limit	initially 4
τ_i^S	system tenure limit	initially 4
$a_{i,t}^P$	current pool-tenure age	staggered
V_t	vacancies before period t	tenure-generated
F_A^{entry}	entrant ability distribution	empirically informed
π_t	assignment order	random permutation

19. Intentionally Absent from the Baseline

The initial model does **not** include:

- an exogenous team target such as T_j^*;
- a BA/Quayle-style popularity term $f(k_j)$;
- independent team prestige attraction;
- restricted opportunity sets;
- heterogeneous team capacities;
- centroid recalculation during an assignment round;
- random attrition beyond tenure-based exit;
- transfers in the initial MBB paradigm;
- heterogeneous tenure;
- heterogeneous ρ.

The minimal baseline is:

$$\boxed{\text{lagged endogenous centroid} + \text{similarity attraction } (\rho) + \text{finite capacity} + \text{staggered turnover.}}$$

20. Important Conceptual Distinctions

Centroid is not a target. $\mu_{j,t}$ is generated by prior composition rather than independently assigned to a team.

ρ is not realized assortment. ρ controls preference strength; realized sorting is an outcome.

Capacity is not similarity. C_j determines availability, not compatibility.

Initialization is not a formation parameter. Empirical $\mu_{j,0}$ values start the system but do not permanently define teams.

Centroid is not Bianconi–Barabási fitness. Fitness is an intrinsic competitive node property in that model; the present centroid is an endogenous summary of prior membership.

21. Extensions Deliberately Deferred

Important later extensions include:

- random or performance-dependent attrition;
- transfers;
- separate pool and system tenure processes;
- Army-style reassignment after pool tenure;

- heterogeneous C_j;
- heterogeneous τ_i;
- restricted opportunity sets;
- alternative similarity kernels;
- multidimensional attributes and centroids;
- team- or agent-specific ρ;
- synthetic initialization;
- empirical calibration of ρ;
- measures of realized bipartite assortment;
- performance feedback mechanisms.

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22. Immediate Open Questions

1. Which empirical player-performance distribution should define the initial F_A^{entry}?
2. Should entrant ability use empirical resampling, a fitted distribution, or another empirically anchored method?
3. What normalization of A is appropriate if retaining $x_{ij,t} = 1 - d_{ij,t}$?
4. Is explicit $x_{ij,t}$ needed computationally, or is $e^{-\rho d_{ij,t}}$ cleaner?
5. What statistic should measure **realized bipartite assortment** while remaining distinct from ρ?
6. How sensitive are outcomes to random sequential assignment under finite capacity?
7. Does lagged-centroid feedback generate persistent stratification, convergence, polarization, or other emergent regimes?

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23. Five-Year Memory

> ***Agents assortatively affiliate with finite-capacity pools according to similarity to each pool's lagged endogenous centroid.***

$$\boxed{\text{yesterday's roster} \rightarrow \text{today's centroid signal} \rightarrow \text{today's sorting} \rightarrow \text{tomorrow's centroid signal.}}$$

Keep the model's objects separate:

$$\boxed{\text{centroid} = \text{state}, \qquad \rho = \text{preference}, \qquad C = \text{constraint}, \qquad \text{realized assortment} = \text{outcome.}}$$

Reference Note

Bianconi, G., & Barabási, A.-L. (2001). *Competition and multiscaling in evolving networks*. Europhysics Letters, 54(4), 436–442. https://doi.org/10.1209/epl/i2001-00260-6